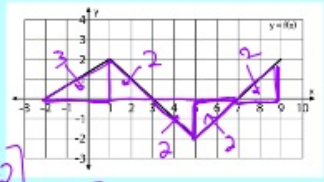
FUN-6.A.1 In some cases, a definite integral can be evaluated by using geometry and the connection between the definite integral and area.
FUN-6.A.2 Properties of definite integrals include the integral of a constant times a function, the integral of the sum of two functions, reversal of limits of integration, and the integral of a function over adjacent intervals.
FUN-6.A.3 The definition of the definite integral may be extended to functions with removable or jump discontinuities.

6-6 Properties of Definite Integrals

Get Ready...

The graph of $f(x)$ consists of three linear segments and is shown in the figure at the right. Evaluate each integral below.

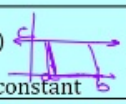


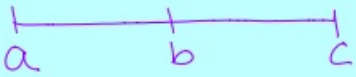
A.) $\int_1^5 f(x)\,dx = 2 - 2 = 0$

B.) $\int_3^{-2} f(x)\,dx = -\int_{-2}^{3} f(x)dx = -[3+2] = -5$

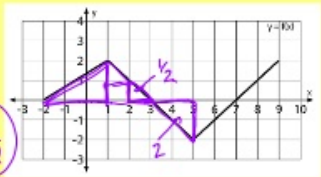
C.) $\int_9^{-2} f(x)\,dx = -\int_{-2}^{9} f(x)dx = -[5-4+2] = -3$

We've talked about the first two properties of integrals below, but there are a few other properties you need to know!

Properties of Integrals	
(1) $\int_a^a f(x)\,dx = 0$	(2) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
(3) $\int_a^b c\,dx = c(b-a)$ where c is any constant 	(4) $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$
(5) Addition rule: $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$	(6) Subtraction rule: $\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$
(7) If $a < b < c$, then $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$	



Example 1: Use the graph of $f(x)$ in the "get ready" section to answer the following questions:



(a) $\int_2^5 3f(x)\,dx = 3\int_2^5 f(x)\,dx$

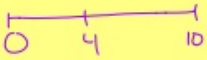
$= 3\left(\frac{1}{2} - 2\right) = 3\left(\frac{-3}{2}\right) = \frac{-9}{2}$

(b) $\int_2^{-2} -4f(x)\,dx = +4\int_{-2}^{2} f(x)\,dx$

$= 4[3 + 1.5] = 18$

Example 2: Given the information above about $g(x)$, find the following:

$$\int_0^{10} g(x)\,dx = 16 \text{ and } \int_4^{10} g(x)\,dx = -4$$



(a) $\int_0^4 g(x)\ dx$

$$\int_0^4 g(x)dx + \int_4^{10} g(x)dx = \int_0^{10} g(x)dx$$

$$\int_0^4 g(x)dx + -4 = 16$$

$$\int_0^4 g(x)dx = 20$$

Example 2: Given the information above about $g(x)$, find the following:

$$\int_0^{10} g(x)\,dx = 16 \text{ and } \int_4^{10} g(x)\,dx = -4$$

(b) $\int_4^0 3g(x)\,dx = -3\int_0^4 g(x)dx = -3(20) = -60$

Example 3: Given that $\int_{-2}^{1} f(x)\,dx = 4$, $\int_{1}^{5} f(x)\,dx = -3$,

and $\int_{-2}^{1} g(x)\,dx = 8$, find the following:

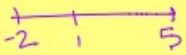
(a) $\int_{-2}^{1} [f(x) + 3]\,dx = \int_{-2}^{1} f(x)dx + \int_{-2}^{1} 3dx$

$= 4 + 3(1 + 2)$

$= 4 + 9 = 13$

Example 3: Given that $\int_{-2}^{1} f(x)\,dx = 4$, $\int_{1}^{5} f(x)\,dx = -3$,

and $\int_{-2}^{1} g(x)\,dx = 8$, find the following:



(b) $\int_{-2}^{5} f(x)\,dx = \int_{-2}^{1} f(x)dx + \int_{1}^{5} f(x)dx$

$= 4 + -3 = 1$

Example 3: Given that $\int_{-2}^{1} f(x)\,dx = 4$, $\int_{1}^{5} f(x)\,dx = -3$,

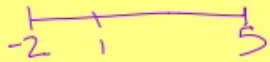
and $\int_{-2}^{1} g(x)\,dx = 8$, find the following:

(c) $\int_{-2}^{1} [f(x) + \underline{2g(x)}]dx = \int_{-2}^{1} f(x)dx + 2\int_{-2}^{1} g(x)dx$

$= 4 + 2(8) = 20$

Example 3: Given that $\int_{-2}^{1} f(x)\,dx = 4$, $\int_{1}^{5} f(x)\,dx = -3$,

and $\int_{-2}^{1} g(x)\,dx = 8$, find the following:



(d) $\int_{0}^{1} f(x)\,dx$ Cannot be determined

Example 3: Given that $\int_{-2}^{1} f(x)\,dx = 4$, $\int_{1}^{5} f(x)\,dx = -3$, and $\int_{-2}^{1} g(x)\,dx = 8$, find the following:

(e) $\int_{-2}^{1} [3f(x) - g(x)]dx$ $= 3\int_{-2}^{1} f(x)dx - \int_{-2}^{1} g(x)dx$

$= 3[4] - 8 = 4$

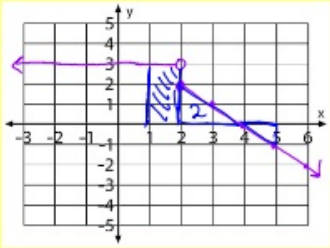
Piecewise Functions and Integrals

Example 4:

The function g is defined by

$$g(x) = \begin{cases} 3 & \text{for } x < 2 \\ 4 - x & \text{for } x \geq 2 \end{cases}$$

What is the value of $\int_1^5 g(x)dx$?

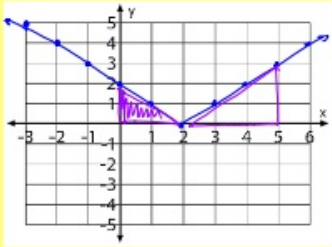


$$\int_1^5 g(x)dx = 3 + 2 - \frac{1}{2}$$

$$= \frac{9}{2} \text{ or } 4.5$$

Example 5:

What is the value of $\int_0^5 |x-2| \; dx$?

$$\int_0^5 |x-2|\,dx = 2 + \frac{9}{2}$$

$$= \frac{13}{2}$$